

Scalability & Future Plan

Date	04 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Deploy the application on a cloud platform (Render/AWS/Azure) for online access.	3–6 Months	Improve accessibility, availability, and support multiple users.
Phase 3	Integrate real-time weather APIs, interactive dashboards, and flood trend visualization.	6–9 Months	Provide more accurate predictions and better analytical insights.
Phase 4	Enhance prediction accuracy using advanced machine learning/deep learning models and automatic dataset updates.	9–12 Months	Improve prediction performance, scalability, and long-term usability.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	The application runs only on a local Flask server.	Limits accessibility for remote users.	High
2	The model uses a static flood dataset and does not automatically update with new weather data.	Prediction accuracy may decrease over time.	High
3	The application predicts flood risk for a single input only and lacks GIS map visualization and live alerts.	Limits advanced analysis and emergency response capabilities.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports users on the local Flask development server.	Deploy on Render, Azure, or AWS to support multiple concurrent users.
2	Data Storage	Uses a local CSV dataset and trained model (flood_model.pkl).	Integrate MySQL/PostgreSQL and automate dataset updates from trusted weather sources.
3	Performance	Suitable for a limited number of prediction requests.	Optimize model loading, implement caching, and improve backend performance.
4	Security	Basic input validation is implemented.	Add user authentication, HTTPS, API security, and secure cloud deployment.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 5	Add SMS/Email notifications and mobile application support.	12–15 Months	Enable instant flood alerts and improve user accessibility.
Phase 6	Integrate IoT sensors and GIS-based flood mapping.	15–18 Months	Provide real-time monitoring, location-based predictions, and improved disaster management.